

**Module 4 Regularization Methods for Predicting College Graduation Rates: Ridge,
LASSO, and Stepwise Regression**

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Introduction:

The idea of predicting student graduation rates is a central concern for college administrators, policymakers, and prospective students who must choose among hundreds of institutions all competing against one another. A student's graduation rate is influenced by a wide array of institutional characteristics which include financial resources, student body composition, faculty support, and admissions selectivity. Many of these predictors are correlated with one another. When ordinary least squares (OLS) regression is applied to a dataset with many correlated predictors, the resulting model is prone to overfitting and also producing highly variable or unstable variable coefficient estimates. Regularization techniques address this problem by adding a penalty term to the loss function that shrinks coefficient estimates toward zero, trading a small amount of bias for a meaningful reduction in variance. The purpose of this report is to build and compare three regression approaches for prediction Grad.Rate (the six-year graduation rate) using the College dataset from the ISLR library – Ridge regression, LASSO (Least Absolute Shrinkage and Selection Operator) regression, and stepwise subset selection. The College dataset contains 777 observations of US colleges and universities and 18 variables describing enrollment, admissions, cost, faculty credentials, and outcomes. The primary questions of this analysis were how do the optimal regularization parameters ($\lambda_{\min}$ and λ_{1se}) selected by cross-validation differ between Ridge and LASSO, and what does that difference imply about the underlying data? They also asked which predictors survive LASSO's variable-selection property and are the surviving predictors intuitive given what is known about higher education finance and admissions? Lastly, they asked how do Ridge, LASSO and stepwise selection compare in terms of predictive accuracy measured by root means square error (RMSE) on held-out test data, and does the model show evidence of overfitting? To answer these

questions, the College dataset was randomly partitioned into a training set with 70% of observations and a test set with 30% of observations. All regularized models were fit using the glmnet package in R and model performance was then evaluated by comparing training and test RMSE which allows for a direct assessment of each model's tendency to overfit the training data. Stepwise selection performed with the base R step () function using both forward and backward steps provided a non-penalized point of comparison so that the added value of regularization could be assessed directly.

Analysis:

The college dataset was loaded, and the private variable was coded as a factor with two levels of Yes and No which indicates whether the institution is a private or public college. The response variable for all models is Grad.Rate which is the percentage of students who graduate. The remaining 16 variables which were Apps, Accept, Enroll, Top10perc, Top25perc, F.Undergrad, P.Undergrad, Outstate, Room.Board, Books, Personal, PhD, Terminal, S.F Ratio, perc.alumni, and expend were used as predictors. The sample () function was used to randomly select 70% of the 777 observations (543 colleges) for the training set with the remaining 234 colleges reserved for the test set. A seed value (set.seed (123)) was used throughout the analysis to ensure the results are reproducible. Rather than relying on R's generic summary () output, descriptive statistics for each numeric variable were computed and expored into a formatted table. The table below reports the mean, standard deviation, minimum, median, and maximum for each of the 16 numeric predictors and the response variable Grad.Rate.

Variable	Mean	SD	Min	Median	Max
Apps	3001.64	3870.2	81	1558	48094

Variable	Mean	SD	Min	Median	Max
Accept	2018.8	2451.11	72	1110	26330
Enroll	779.97	929.18	35	434	6392
Top10perc	27.56	17.64	1	23	96
Top25perc	55.8	19.8	9	54	100
F.Undergrad	3699.91	4850.42	139	1707	31643
P.Undergrad	855.3	1522.43	1	353	21836
Outstate	10440.67	4023.02	2340	9990	21700
Room.Board	4357.53	1096.7	1780	4200	8124
Books	549.38	165.11	96	500	2340
Personal	1340.64	677.07	250	1200	6800
PhD	72.66	16.33	8	75	103
Terminal	79.7	14.72	24	82	100
S.F.Ratio	14.09	3.96	2.5	13.6	39.8
perc.alumni	22.74	12.39	0	21	64
Expend	9660.17	5221.77	3186	8377	56233
Grad.Rate	65.46	17.18	10	65	118

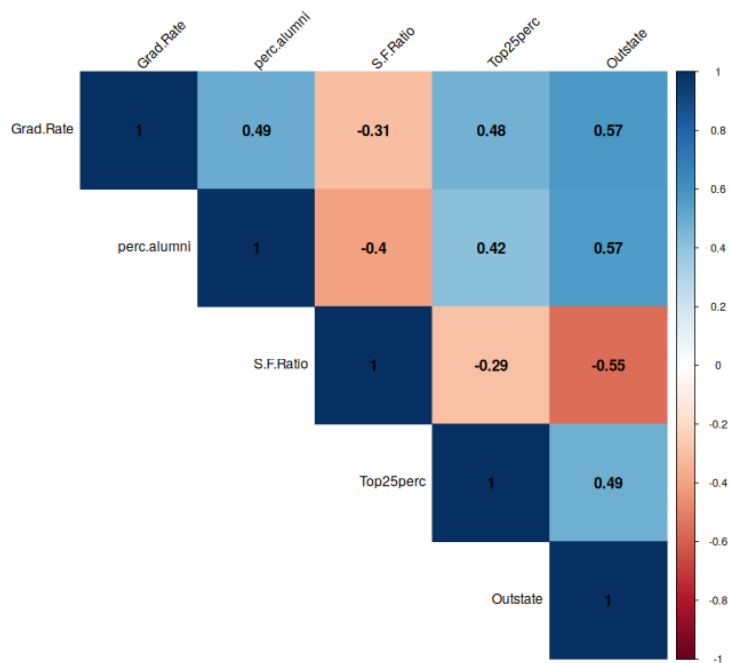
The wide range in scale across variables (Apps ranges from 81 – 48,094, while S.F.Ratio ranges from 2.5 – 39.8) illustrates a large benefit of Ridge and LASSO regression. Both of these are

estimated using standardized predictors internally so that variables measured on very different scales can be penalized fairly.

The correlation matrix presented here has five variables and they are the response Grad.Rate, and the four predictors that emerged as most influential across the Ridge, LASSO, and stepwise models (perc.alumni, S.F.Ratio, Top25perc, and Outstate).

Variable	Grad.Rate	perc.alumni	S.F.Ratio	Top25perc	Outstate
Grad.Rate	1	0.49	-0.31	0.48	0.57
perc.alumni	0.49	1	-0.4	0.42	0.57
S.F.Ratio	-0.31	-0.4	1	-0.29	-0.55
Top25perc	0.48	0.42	-0.29	1	0.49
Outstate	0.57	0.57	-0.55	0.49	1

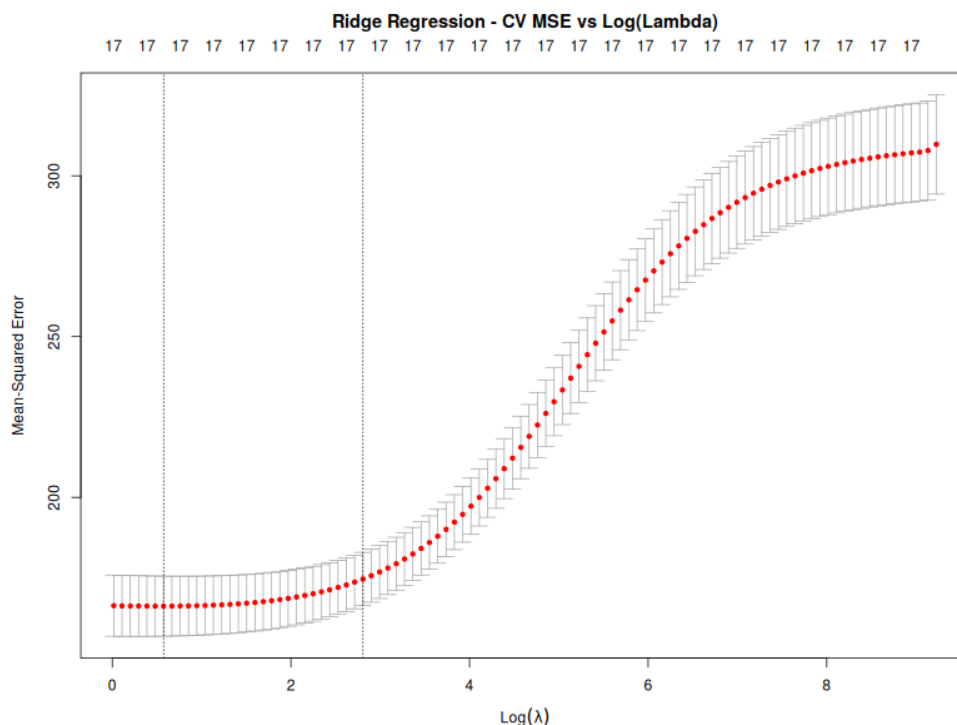
Correlation Matrix: Grad.Rate and Top Predictors



Grad.Rate is most strongly correlated with Outstate ($r = 0.57$) and perc.alumni ($r = 0.49$), and is moderately correlated with Top25perc (0.48). S.F.Ratio is negatively correlated with Grad.Rate ($r = -0.31$) and negatively correlated with all four of the other variables. This is consistent with the idea that colleges with lower student-to-faculty ratios tend to be wealthier, more selective institutions with higher alumni engagement and higher graduation rates. None of the pairwise correlations among predictors exceed 0.6 in absolute value which suggests that while these variables share some overlapping information, multicollinearity among them is only moderate rather than severe. This is consistent with the earlier finding that Ride, LASSO, and stepwise selection all produced fairly similar, stable results.

Ridge regression applies an L2 penalty to the regression coefficients which shrinks them toward zero but never sets them exactly equal to zero. This makes Ridge well suited to situations where most or all predictors are believed to contribute at least a small amount of explanatory power. The `cv.glmnet` function was used with $\alpha = 0$ (the Ridge setting) to perform 10-fold cross-validation across a grid of lambda values, choosing the value that minimizes cross-validated mean squared error (MSE). The cross-validation procedure identified a `lambda.min` of 1.78 and a `lambda.1se` of 16.56. The `lambda.min` value of the tuning parameter that produced the lowest cross-validated MSE, while `lambda.1se` is the largest value of lambda within one standard error of the minimum. The 1se value is intentionally larger than `lambda.min` because it selects a more heavily regularized (simpler, more conservative) model that is statistically indistinguishable from the best-performing model but is less likely to overfit the training data. In this case, the 1se lambda is roughly 9 times larger than `lambda.min` which indicates that a wide range of penalty strengths produce statistically similar cross-validated performance, and that the

cross-validation MSE curve is relatively flat near its minimum. Since the goal of the analysis was predictive accuracy, $\lambda_{\min}$ was used to fit the final Ridge model.



This figure plots the cross-validated MSE (with error bars representing one standard error) against the log of the lambda. The left-most dashed vertical line marks $\log(\lambda_{\min})$ and the right-most dashed line markets $\log(\lambda_{1se})$. The numbers along the top of the plot show that all 17 predictors remain in the model at every value of lambda shown, which is expected because Ridge regression does not perform variable selection. The U-shape of the curve shows that MSE is high when lambda is very small (the model is essentially unpenalized and overfits) and rises again as lambda grows very large (the model becomes too simple, and coefficients are shrunk too much which underfits the data). The relatively flat bottom of the curve between roughly $\log(\lambda) = 0$ and $\log(\lambda) = 2.5$ indicates that Ridge regression's predictive performance is fairly insensitive to the exact choice of lambda within that range.

Unlike LASSO, none of the Ridge coefficients are exactly zero, since the L2 penalty shrinks coefficients continuously without eliminating them.

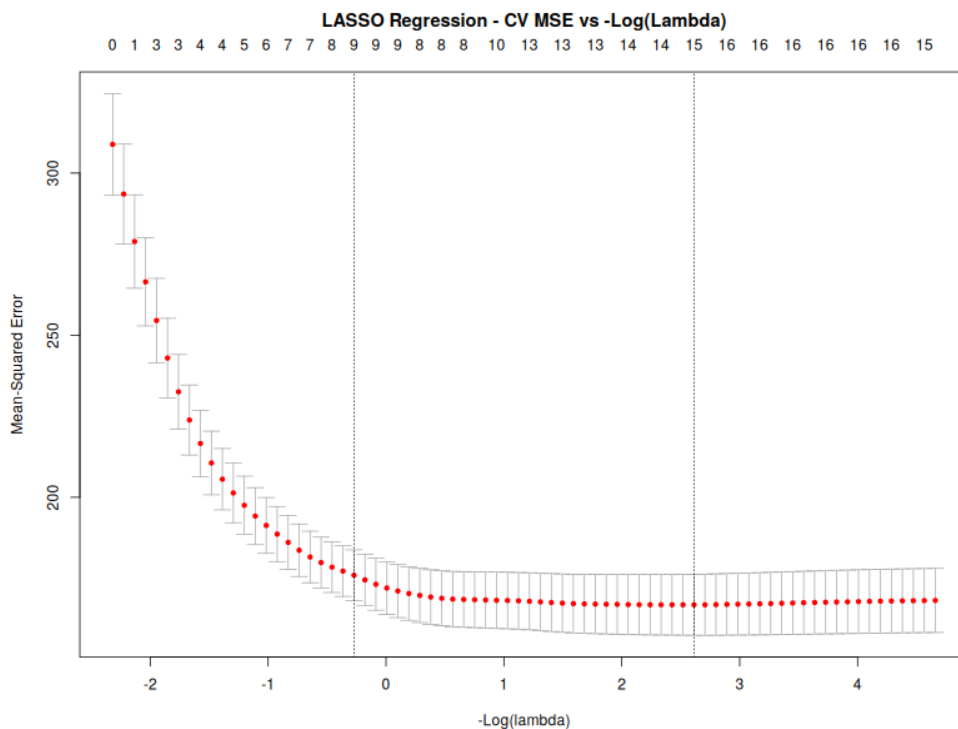
Predictor	Coefficient
Intercept	39.709
PrivateYes	4.332
Apps	0.0008
Accept	0.0002
Enroll	-0.0003
Top10perc	0.0688
Top25perc	0.1391
F.Undergrad	-0.00005
P.Undergrad	-0.0012
Outstate	0.0006
Room.Board	0.0020
Books	-0.0005
Personal	-0.0029
PhD	0.0943
Terminal	-0.0876
S.F.Ratio	-0.1760
perc.alumni	0.2908

Predictor	Coefficient
Expend	-0.0003

The largest-magnitude coefficients belong to PrivateYes (4.33), perc.alumni (0.29), S.F.Ratio (-0.18), Top25perc (0.14), PhD (0.09), and Terminal (-0.09). This suggests, holding other variables constant, private institutions and colleges with a higher percentage of alumni who donate, tend to have higher graduations rates. On the flip side, a higher student-to-faculty ratio is associated with lower graduation rates. The coefficients for Terminal (percent of faculty with a terminal degree) and PhD (percent of faculty with a PhD) have opposite signs which is counterintuitive. This pattern likely reflects collinearity between these two closely related variables which is a problem that Ridge regression is specifically designed to stabilize by shrinking correlated coefficients toward each other rather than allowing either one to take an extreme, unstable value. The RMSE of the Ridge model was calculated on both the training and test sets using $\sqrt{\text{mean}((\text{actual} - \text{predicted})^2)}$ - The training RMSE was 12.54 and the test RMSE was 13.02. Because the test RMSE is only slightly higher than the training RMSE (a difference of about 0.48 percentage points of graduation rate), the Ridge model does not show meaningful evidence of overfitting. It generalizes to unseen colleges nearly as well as it fits the colleges it was trained on.

LASSO regression applies an L1 penalty to the coefficients, which has the important property of shrinking some coefficients all the way to exactly zero. This gives LASSO an automatic variable-selection capability that Ridge regression lacks. The `cv.glmnet()` function was used with $\alpha = 1$ for the LASSO setting in order to select the optimal λ via 10-fold cross-validation. Cross-validation identified a λ_{min} of 0.073 and a $\lambda_{1\text{se}}$ of 1.312

and as with Ridge, λ_{1se} is substantially larger than λ_{min} (roughly 18 times larger). This reflects a fairly flat region of the cross-validation curve near its minimum. However, because LASSO's penalty acts on the absolute value of the coefficients rather than their square, even a modest increase in λ causes additional coefficients to be shrunk to exactly zero. At λ_{min} , `cy.glmnet` reports 15 non-zero coefficients (of 17 possible predictors, not counting the intercept), while at λ_{1se} only 9 predictors remain in the model. This is a key conceptual difference from Ridge because increasing λ for LASSO actively simplifies the model by removing predictors, not merely shrinking them.



This figure shows the same general U-shape as the Ridge plot, but the numbers along the top of the plot (representing the number of non-zero coefficients retained at each λ) increase steadily from 0 on the far left to 17 on the far right. This illustrates LASSO's progressive variable elimination as λ increases, moving from right to left. The left-most dashed line

marks $-\log(\lambda_{1se})$ and the right-most dashed line marks $\log(\lambda_{min})$. Since the curve is very flat between these two points, a simpler model with roughly half as many predictors achieves essentially the same cross-validated MSE as the more complex λ_{min} model. This supports using LASSO for both prediction and for identifying a smaller, more interpretable subset of important variables.

The LASSO model was fit at λ_{min} for consistency with the Ridge analysis and the resulting coefficients are shown in a table below with a period (.) indicating a coefficient shrunk exactly to zero.

Predictor	Coefficient
Intercept	38.772
PrivateYes	4.669
Apps	0.0013
Accept	-0.0002
Enroll	-0.0007
Top10perc	0.0109
Top25perc	0.1776
F.Undergrad	0 (removed)
P.Undergrad	-0.0013
Outstate	0.0007
Room.Board	0.0020

Predictor	Coefficient
Books	0 (removed)
Personal	-0.0029
PhD	0.1247
Terminal	-0.1278
S.F.Ratio	-0.1792
perc.alumni	0.3232
Expend	-0.0004

Two coefficients, F.Undergrad (number of full-time undergraduates) and Books (estimated book costs), were reduced exactly to zero and removed from the model, while all other predictors were retained. This makes sense because the number of full-time undergraduates is likely redundant with other size-related variables in the model such as Apps, Accept, and Enroll – estimated book costs plausibly have little direct relationship with whether a student ultimately graduates. The remaining, non-zero coefficients follow a very similar pattern to the Ridge coefficients in both sign and relative magnitude. PrivateYes, perc.alumni, S.F.Ratio, Top25perc, PhD, and Terminal stand out as the strongest predictors. The RMSE of the LASSO model was 12.49 on the training set and 13.17 on the test set. The gap between training and test RMSE (about 0.68 percentage points) is small which indicates that the LASSO model also generalizes well and is not substantially overfit.

To provide a point of comparison, a stepwise selection model was also fit. Starting from the full model (Grad.Rate regressed on all 17 predictors), the step () function was run with

direction = “both”, which allowed variables to be both added and removed at each iteration based on AIC. The stepwise procedure retained 12 predictors, which meant dropping Enroll, F.Undergrad, Books, S.F.Ratio, and Top10perc from the full model. It is interesting that the stepwise model dropped S.F.Ratio because both Ridge and LASSO retained it as one of the largest-magnitude and most important predictors. This demonstrates that stepwise selection based on individual variable significance and AIC doesn’t always agree with regularization-based selection, particularly when predictors are correlated with one another.

Predictor	Estimate	Std. Error	t value	p value
(Intercept)	35.0093	3.9356	8.8956	0
PrivateYes	5.4254	2.0151	2.6923	0.0073
Apps	0.0017	5e-04	3.3268	9e-04
Accept	-0.0011	8e-04	-1.4687	0.1425
Top25perc	0.1788	0.0383	4.6689	0
P.Undergrad	-0.0013	4e-04	-3.1615	0.0017
Outstate	8e-04	3e-04	2.942	0.0034
Room.Board	0.0021	7e-04	3.166	0.0016
Personal	-0.0029	9e-04	-3.2891	0.0011
PhD	0.1518	0.0669	2.2678	0.0237
Terminal	-0.1536	0.0723	-2.1241	0.0341
perc.alumni	0.3311	0.0567	5.8345	0

Predictor	Estimate	Std. Error	t value	p value
Expend	-5e-04	2e-04	-3.0732	0.0022

This table reports the exported coefficients for the stepwise model, including standard errors, t-values, and p-values for each retained predictor. All three models produced very similar predictive performance with test RMSE ranging narrowly from 13.02 (Ridge) to 13.17 (LASSO). Ridge regression achieved the lowest test RMSE, narrowly outperforming both LASSO and stepwise selection, while LASSO had the lowest training RMSE. The result makes sense because Ridge shrinks all coefficients rather than eliminating any of them, so it tends to perform comparably or slight better than LASSO – this occurs when most predictors genuinely contribute more (even if a small amount) signal to the response, which appears to be the case here given that LASSO removed only two of the seventeen predictors. This outcome was largely expected because the College dataset predictors are moderately correlated, but each plausibly carries at least some independent relationship with graduation rate. This favors Ridge’s shrink everything a little approach over LASSO’s eliminate the irrelevant, shrink the rest approach. Because the stepwise model performed nearly identically to both regularized models, this suggests for this dataset and train/test split, the College data does not appear to contain a large number of highly noisy or redundant predictors that only regularization could suppress. LASSO is preferred for reporting and interpretation purposes because it automatically produces a simpler, more interpretable model by zeroing out the least useful predictors.

Conclusion:

This analysis used Ridge regression, LASSO regression, and stepwise selection to predict college graduation rates using the College dataset. Across all three modeling approaches, the percentage of alumni who donate (perc.alumni), student-faculty ratio (S.F.Ratio), whether the institution is private (Private), and the percentage of students in the top 25% of their high school class (Top25perc) consistently emerged as the strongest predictors of graduation rate. Since the difference in predictive accuracy between Ridge and LASSO methods is small, the choice between them in practice should be guided by the analyst's goals. LASSO is preferable when interpretability and a reduced feature set are priorities, whereas Ridge is preferable when maximizing predictive accuracy is the objective and there is reason to believe most predictors carry at least some signal. For a college administrator or policy analyst using this type of model, the recommendation would be the LASSO because it retains nearly all of the predictive power of the complex Ridge model, while offering a simpler set of drivers to communicate and act upon: increasing alumni engagement, managing student-faculty ratios, and maintaining a stronger incoming class profile appear to be associated with higher graduation rates across all three modeling approaches tested in this analysis.

References

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Appendix:

```

1  #Joseph Tarlin, ALY 6015, 07/28/2026, Regularization Assignment Module 4
2  set.seed(123)
3  data(College)
4  College$Private <- as.factor(College$Private)
5  str(College)
6  sum(is.na(College))
7  #Descriptive Statistics
8  numeric_vars <- College[, sapply(College, is.numeric)]
9  desc_stats <- data.frame(Variable = names(numeric_vars),
10 Mean = round(sapply(numeric_vars, mean, na.rm = TRUE), 2),
11 SD = round(sapply(numeric_vars, sd, na.rm = TRUE), 2),
12 Min = round(sapply(numeric_vars, min, na.rm = TRUE), 2),
13 Median = round(sapply(numeric_vars, median, na.rm = TRUE), 2),
14 Max = round(sapply(numeric_vars, max, na.rm = TRUE), 2),
15 row.names = NULL)
16 print(desc_stats)
17 #Correlation matrix with 5 variables
18 corr_vars <- College[, c("Grad.Rate", "perc.alumni", "S.F.Ratio", "Top25perc", "Outstate")]
19 corr_matrix <- round(cor(corr_vars), 2)
20 print(corr_matrix)
21 corrplot(cor(corr_vars), method = "color", type = "upper", addCoef.col = "black",
22 tl.col = "black", tl.srt = 45, number.cex = 1.1,
23 title = "Correlation Matrix: Grad.Rate and Top Predictors", mar = c(0,0,2,0))
24 #Train / test split, 70 / 30
25 trainIndex <- sample(x = nrow(College), size = nrow(College) * 0.7)
26 train <- College[trainIndex, ]
27 test <- College[-trainIndex, ]
28 cat("Train rows:", nrow(train), " Test rows:", nrow(test), "\n")
29 train_x <- model.matrix(Grad.Rate ~ ., data = train)[, -1]
30 test_x <- model.matrix(Grad.Rate ~ ., data = test)[, -1]

31 train_y <- train$Grad.Rate
32 test_y <- test$Grad.Rate
33 #Ridge regression
34 set.seed(123)
35 cv_ridge <- cv.glmnet(train_x, train_y, alpha = 0)
36 print(cv_ridge)
37 ridge_lambda_min <- cv_ridge$lambda.min
38 ridge_lambda_1se <- cv_ridge$lambda.1se
39 cat("Ridge lambda.min:", ridge_lambda_min, "\n")
40 cat("Ridge lambda.1se:", ridge_lambda_1se, "\n")
41 plot(cv_ridge)
42 title("Ridge Regression - CV MSE vs Log(Lambda)", line = 2.5)
43 ridge_model <- glmnet(train_x, train_y, alpha = 0, lambda = ridge_lambda_min)
44 #Predictions
45 ridge_pred_train <- predict(ridge_model, newx = train_x)
46 ridge_pred_test <- predict(ridge_model, newx = test_x)
47 ridge_rmse_train <- sqrt(mean((train_y - ridge_pred_train)^2))
48 ridge_rmse_test <- sqrt(mean((test_y - ridge_pred_test)^2))
49 cat("Ridge RMSE (train):", ridge_rmse_train, "\n")
50 cat("Ridge RMSE (test):", ridge_rmse_test, "\n")
51 #Lasso regression
52 set.seed(123)
53 cv_lasso <- cv.glmnet(train_x, train_y, alpha = 1)
54 print(cv_lasso)
55 lasso_lambda_min <- cv_lasso$lambda.min
56 lasso_lambda_1se <- cv_lasso$lambda.1se
57 cat("LASSO lambda.min:", lasso_lambda_min, "\n")
58 cat("LASSO lambda.1se:", lasso_lambda_1se, "\n")
59 plot(cv_lasso)
60 title("LASSO Regression - CV MSE vs Log(Lambda)", line = 2.5)
61 lasso_model <- glmnet(train_x, train_y, alpha = 1, lambda = lasso_lambda_min)
62 print(coef(lasso_model))

```

```

63 lasso_pred_train <- predict(lasso_model, newx = train_x)
64 lasso_pred_test <- predict(lasso_model, newx = test_x)
65 lasso_rmse_train <- sqrt(mean((train_y - lasso_pred_train)^2))
66 lasso_rmse_test <- sqrt(mean((test_y - lasso_pred_test)^2))
67 cat("LASSO RMSE (train):", lasso_rmse_train, "\n")
68 cat("LASSO RMSE (test):", lasso_rmse_test, "\n")
69 #Stepwise selection comparison
70 full_model <- lm(Grad.Rate ~ ., data = train)
71 step_model <- step(full_model, direction = "both", trace = 0)
72 cat("Stepwise formula:\n")
73 print(formula(step_model))
74 #Exporting stepwise coefficient table
75 step_coef_table <- as.data.frame(coef(summary(step_model)))
76 step_coef_table <- data.frame(Predictor = rownames(step_coef_table), round(step_coef_table, 4))
77 rownames(step_coef_table) <- NULL
78 names(step_coef_table) <- c("Predictor", "Estimate", "Std.Error", "t.value", "p.value")
79 print(step_coef_table)
80 cat("Stepwise Adjusted R-squared:", summary(step_model)$adj.r.squared, "\n")
81 step_pred_train <- predict(step_model, newdata = train)
82 step_pred_test <- predict(step_model, newdata = test)
83 step_rmse_train <- sqrt(mean((train_y - step_pred_train)^2))
84 step_rmse_test <- sqrt(mean((test_y - step_pred_test)^2))
85 cat("Stepwise RMSE (train):", step_rmse_train, "\n")
86 cat("Stepwise RMSE (test):", step_rmse_test, "\n")
87 #Summary table
88 results <- data.frame(
89   Model = c("Ridge", "LASSO", "Stepwise"),
90   Lambda_min = c(ridge_lambda_min, lasso_lambda_min, NA),
91   RMSE_Train = c(ridge_rmse_train, lasso_rmse_train, step_rmse_train),
92   RMSE_Test = c(ridge_rmse_test, lasso_rmse_test, step_rmse_test))
93 print(results)
94 ridge_coefs <- as.data.frame(as.matrix(coef(ridge_model)))
95 lasso_coefs <- as.data.frame(as.matrix(coef(lasso_model)))
96 write.csv(ridge_coefs, "ridge_coefs.csv")
97 write.csv(lasso_coefs, "lasso_coefs.csv")

```